

Malla Harshal Kumar

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Summary

Final-year Computer Science undergraduate (VIT-AP, CGPA 8.33) with strong fundamentals in data structures, algorithms, OOP, and DBMS in Java and Python, plus hands-on experience with concurrent, real-time systems. Delivered a full-stack, database-backed platform end-to-end (React, Node.js, MongoDB) with secure authentication and transactional workflows, and an autonomous closed-loop AI control system pairing physics-based simulation with local LLM agents. Test-first and debugging-driven, with operating-systems and concurrency grounding. Seeking a Technology Analyst role in UBS's Group Technology Office to build resilient, high-integrity systems for global banking and markets technology.

Skills

Languages: Java, Python, C, SQL, JavaScript

AI/ML: Generative AI, foundation models, prompt engineering, agentic tool-calling, Model Context Protocol (MCP), Quantum Computing (Qiskit, PennyLane)

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Concurrency & Multithreading (RTOS scheduling)

Backend & Web: FastAPI, Node.js, REST APIs, React, HTML, CSS, real-time systems (Socket.io), analytics dashboards (Recharts)

Databases: SQL (relational modelling, queries), MongoDB (NoSQL)

Tools & Cloud: Git, VS Code, IBM Watsonx.ai, cloud deployment

Competencies: Problem Solving, Team Leadership, Time Management, Communication

Projects

Eco-Loop — Closed-Loop AI Building Energy Controller

2026

Python, FastAPI, EnergyPlus, Model Context Protocol (MCP), Ollama (local LLM)

- Built a **closed-loop, physics-in-the-loop control system** pairing an EnergyPlus building-energy simulation with a locally-hosted open-source LLM (Ollama/Llama 3.1), exchanging observations and control actions through a custom **MCP tool-calling server**.
- Achieved a verified **6.25% reduction in HVAC electricity use** (232.4 kWh to 217.8 kWh) across a 24-hour real EnergyPlus run while holding occupied thermal comfort within Predicted Mean Vote (PMV) ± 0.5 bounds (max |PMV| 0.268).
- Designed a **deterministic, Pydantic-validated safety layer** that gates every LLM-proposed setpoint before actuation, guaranteeing zero unsafe actuator writes across 50 MCP calls and 24 real-time hourly decisions; verified with an automated pytest suite covering the API, MCP handshake, and comfort validator.

Bay Manager — Garage Service Management Platform

2025

Full-stack MERN application (React, Node.js/Express, MongoDB)

- Built a full-stack MERN platform with **JWT-based authentication** and role-based access control across Admin, Staff, and Customer roles.
- Designed REST APIs and MongoDB schemas covering bookings, job cards, invoices, payments, and feedback, with a live **analytics dashboard** (Recharts).
- Implemented **real-time job-status notifications** with Socket.io, integrated Stripe for invoice payments, and secured credentials using bcrypt password hashing.

Education

VIT-AP University, Amaravati, India

2023 – 2027 (Expected)

B.Tech in Computer Science Engineering | CGPA: 8.33

Shiv Jyoti Sr. Secondary School (CBSE), Kota

2020 – 2022

Intermediate (Class 12) | 79.4%

Gowtham Concept School (CBSE), Shantinagar

2020

Class 10 | 87.6%

Certifications & Continuous Learning

Real-Time Operating Systems (NPTEL, IIT Kharagpur) — RTOS design & scheduling, Feb–Mar 2026

IBM Certified Associate — Generative AI with Watsonx

AWS Academy Graduate — Cloud Foundations & Cloud Architecting (Amazon Web Services), Sep–Nov 2025

Quantum Computing Program — WISER / Amaravati Quantum Valley, Qubitech (Qiskit, qBraid, PennyLane), 2025–2026